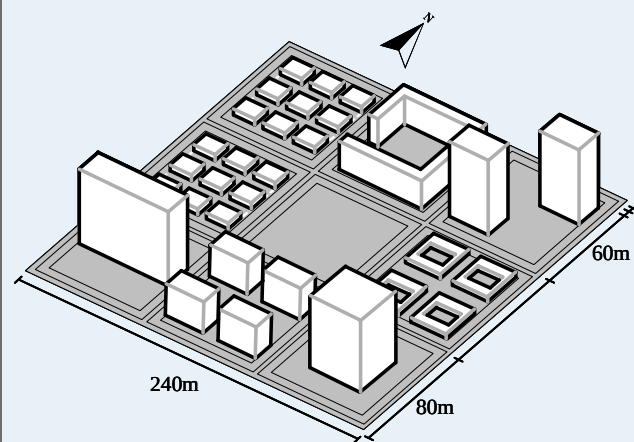


1. Feasible morphology generation



Randomly sampled

- open-space position
- building prototype
- number of floors
- block orientation θ

Feasible block geometry $\rightarrow$ 12 morphology descriptors

2. Analytic response and surrogate modeling

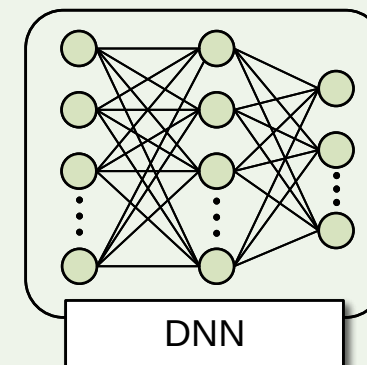
12 morphology descriptors



Analytic response generator



EUI_t / EG / H



DNN

Nested sample pools

- 500 blocks
- 1000 blocks
- 1500 blocks
- 2000 blocks

Selected DNN surrogate

Inputs: 12 descriptors
Outputs: EUI_t, EG, H

4. Feasible design assessment

Descriptor-space candidate



Nearest generated feasible block



Mapping distance

Mapping outcomes

- unique feasible blocks
- many-to-one candidate mapping

Cross-model assessment

- 18 directly sampled blocks
- 6 optimizer-linked blocks
- EUI_t / EG proxy / H

Climate sensitivity

- 4 representative blocks
- Dongtai baseline
- Beijing / Guangzhou / Harbin

3. Surrogate-assisted optimization

Shared surrogate environment

Selected DNN + guardrail + output clipping

DDPG

- 20 seeds per setting
- 600 episodes per seed
- 40 queries per episode

NSGA-II

- 20 independent runs
- 24,000 queries per run

$$z_i = \frac{y_i - y_i^{\min}}{y_i^{\max} - y_i^{\min}}$$

$$u = (0, 1, 1)$$

$$d_w = \frac{\|w \odot (z - u)\|_2}{\|w\|_2}, \quad R = 1 - d_w$$

w : axis-scaling coefficients

Additional comparison

- CMA-ES
- RandomSearch

selected surrogate

candidate solutions